Lab: Using Archive Utilities on Linux

Objective

- Learn how to create and extract archives using common Linux tools.
- Understand how to compress directories and files with different utilities:
    - **zip/unzip**
    - **tar** (with no compression, gzip, bzip2, and xz)
    - **cpio** (combined with gzip for compression)

Prerequisites

- A Linux system with a terminal.
- Installation of utilities (most are pre-installed on many distributions). If needed:

sudo apt-get install zip unzip tar gzip bzip2 xz-utils cpio

Part 1: Preparing a Sample Directory

1. **Create a Working Directory**

Open your terminal and create a new directory named labfiles:

mkdir labfiles

2. **Populate the Directory**

Create several text files and a subdirectory with a file inside. This structure will serve as your sample data.

cd labfiles

echo "This is the first file." > file1.txt

echo "This is the second file." > file2.txt

echo "This is the third file." > file3.txt

mkdir subdir

echo "This file is inside a subdirectory." > subdir/subfile.txt

cd ..

Your directory structure should now look like this:

```
labfiles/
 ├── file1.txt
 ├── file2.txt
 ├── file3.txt
└── subdir/
   └── subfile.txt
```

Part 2: Archiving with zip and unzip

1. **Create a zip Archive**

Archive the entire labfiles directory into a file called labfiles.zip:

zip -r labfiles.zip labfiles

- The -r flag ensures that directories are processed recursively.

2. **Extract the Zip Archive**

Create a new directory to extract the contents and unzip the archive:

mkdir labfiles_zip_extract

unzip labfiles.zip -d labfiles_zip_extract

3. **Verify the Extraction**

Check the labfiles_zip_extract directory to confirm that the file structure matches the original.

Part 3: Creating tar Archives

A. Uncompressed tar Archive

1. **Create an Uncompressed tar Archive**

Use tar to create an archive named labfiles.tar:

tar -cvf labfiles.tar labfiles

- **Flags:**
  - -c: Create a new archive.
  - -v: Verbosely list files processed.
  - -f: Specifies the filename.

2. **Extract the Uncompressed tar Archive**

Create an extraction directory and extract the archive:

mkdir labfiles_tar_extract

tar -xvf labfiles.tar -C labfiles_tar_extract

B. Compressed tar Archive with gzip

1. **Create a gzip Compressed tar Archive**

Create an archive with gzip compression:

tar -czvf labfiles.tar.gz labfiles

- **Flag** -z tells tar to compress the archive using gzip.

2. **Extract the gzip Compressed tar Archive**

Extract into a new directory:

mkdir labfiles_tar_gz_extract

tar -xzvf labfiles.tar.gz -C labfiles_tar_gz_extract

C. Compressed tar Archive with bzip2

1. **Create a bzip2 Compressed tar Archive**

Use the following command:

tar -cjvf labfiles.tar.bz2 labfiles

- **Flag** -j instructs tar to use bzip2 compression.

2. **Extract the bzip2 Compressed tar Archive**

Extract the archive:

mkdir labfiles_tar_bz2_extract

tar -xjvf labfiles.tar.bz2 -C labfiles_tar_bz2_extract

D. Compressed tar Archive with xz

1. **Create an xz Compressed tar Archive**

Create the archive as follows:

tar -cJvf labfiles.tar.xz labfiles

- **Flag** -J tells tar to use xz compression.

2. **Extract the xz Compressed tar Archive**

Extract the archive:

mkdir labfiles_tar_xz_extract

tar -xJvf labfiles.tar.xz -C labfiles_tar_xz_extract

3. **Verify Each Extraction**

Ensure all the extracted directories (labfiles_tar_extract, labfiles_tar_gz_extract, labfiles_tar_bz2_extract, and labfiles_tar_xz_extract) have the same structure as the original labfiles directory.

Part 4: Archiving with cpio (Combined with gzip)

1. **Create a cpio Archive of Specific Files**

Use the find command to list all .txt files in labfiles and create an archive with cpio:

find labfiles -type f -name "*.txt" | cpio -ov > labfiles.cpio

- **Flags for cpio:**
  - -o: Create an archive.
  - -v: Verbose output.

2. **Compress the cpio Archive with gzip**

Compress the archive:

gzip labfiles.cpio

This command produces labfiles.cpio.gz.

3. **Extract the cpio Archive**

Create an extraction directory and extract the gzip-compressed cpio archive:

mkdir labfiles_cpio_extract

gunzip -c labfiles.cpio.gz | (cd labfiles_cpio_extract && cpio -idmv)

- **Notes:**
  - gunzip -c decompresses the archive to standard output.
  - The pipe | passes the decompressed data into cpio for extraction.

- -i: Extract files.
- -d: Create directories as needed.
- -m: Preserve modification times.
- -v: Verbose output.

4. **Verify the cpio Extraction**

Check the labfiles_cpio_extract directory to ensure that all .txt files have been extracted correctly.

Wrap-Up and Clean-Up

- **Review:** Ensure you understand each command and its flags. Verify that the extracted directories match your original labfiles structure.
- **Clean Up (Optional):** Remove temporary archives and extraction directories if needed:

rm -rf labfiles_zip_extract labfiles_tar_extract labfiles_tar_gz_extract labfiles_tar_bz2_extract labfiles_tar_xz_extract labfiles_cpio_extract

rm labfiles.zip labfiles.tar labfiles.tar.gz labfiles.tar.bz2 labfiles.tar.xz labfiles.cpio.gz

- **Extend the Lab:** Experiment with additional options such as excluding files from tar archives (--exclude) or using encryption with zip (-e). Try modifying files in the original directory and re-running backup commands to see how updates are handled.

Conclusion

In this lab, you learned how to:

- Create and extract archives using **zip** and **unzip**.
- Use **tar** in its uncompressed form and combined with different compression tools: **gzip**, **bzip2**, and **xz**.
- Use **cpio** together with **gzip** to archive specific files.